

Nearby cycles cosheaf and derived Satake with $\mathbb{Z}$ -coefficients

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Let G complex reductive group, $k = \mathbb{C}((t))$, $\mathbb{C}[[t]] = \mathcal{O}$

$$G(\mathcal{O}) \backslash G_k = G(k) / G(\mathcal{O}) = \coprod_{\lambda \in X_*(T)^+} G(\mathcal{O}) t^\lambda G(\mathcal{O}) / G(\mathcal{O})$$

$$\lambda: G_m \rightarrow G, \quad t^\lambda = \lambda(t) \in G(k)$$

(Classical) Satake equivalence

$$\text{Pon}^{G(\mathcal{O})}(G_k, D(\mathbb{Z})) \simeq \text{Rep}_{\check{G}, \mathbb{Z}}^{\text{tg.}}$$

mod p ↖ Chevalley model for the Langlands dual group

$$IC_\lambda \longleftrightarrow L_\lambda$$

$$D(\text{Rep}_{\check{G}}) \longrightarrow \text{Shv}^{G(\mathcal{O})}(G_k, D(\mathbb{Z})) =: \text{Sph}_G$$

$$\left(\begin{array}{l} \text{or } \text{IndPerv}(B\check{G}) \\ \text{or } \text{Qcoh}(B\check{G}) \end{array} \right) \xrightarrow{\uparrow} \text{Sph}_G \quad \text{not fully faithful}$$

$$BG(\mathcal{O}) \xrightarrow{i} G(\mathcal{O}) \backslash G_k$$

$$i_* \mathbb{Z} \longleftrightarrow \mathbb{Z} \text{ triv rep. Kempf th.}$$

$$R\Gamma_{\text{Sph}_G}(i_* \mathbb{Z}, i_* \mathbb{Z}) \quad R\Gamma_{\text{D}(\text{Rep}_{\check{G}})}(\mathbb{Z}, \mathbb{Z}) \stackrel{!}{=} \mathbb{Z}$$

$$R\Gamma_{\text{sing}}(BG(\mathcal{O}), \mathbb{Z}) \underset{\text{h.e.}}{\hat{=}} BG(\mathbb{C})$$

Expectation $\text{Sph}_G \xrightarrow{\sim} \text{IndCoh}_{\text{nilp}}(\text{Map}(S^2, B\check{G}))$

$$G(\mathcal{O}) \backslash G_k \underset{\text{h.e.}}{\simeq} \Omega^2 BG \quad \left(e_{\check{G}}^R \right) / \check{G}$$

over $\mathbb{Q}$ is known (Bogomolov - Finkelberg)

Observation. RHS is (close enough to) the $\mathbb{F}_2$ -centre of $\text{IndPerf}(\mathcal{B}_\Lambda^\vee)$

Given $A \rightsquigarrow M$, to produce a functor $A \rightarrow \mathcal{Z}_{\mathbb{F}_2}(M)$,

$\mathbb{F}_2$ -monoidal-cat.

it is enough to lift $A \rightsquigarrow M \in \text{LMod}$ to an $\mathbb{F}_2$ -algebra.

$$(A \rightsquigarrow M) \in \text{Alg}_{\mathbb{F}_2}(\text{LMod})$$

$$\begin{array}{cc} \swarrow & \searrow \\ A \in \text{Alg}_{\mathbb{F}_2} & M \in \text{Alg}_{\mathbb{F}_2} \end{array}$$

Rem. there is a suitable module over Sph_Λ given by Kirillov - Whittaker model

Notation. let X be a stack with G_a -action,

$$\text{Shv}_{\text{exp}}(X) := \text{Shv}(X) / \text{Shv}(X/G_a)$$

$$\begin{array}{ccc} N(K) & \rightsquigarrow & \omega_{K,\Lambda} \\ & \hookrightarrow & \uparrow \\ & \omega_m & \check{\rho} \end{array}$$

$$\begin{array}{ccc} N(K) & \xrightarrow{\chi} & G_a \\ \uparrow & & \\ \ker \chi & & \end{array}$$

$$X = \ker(\chi) \times_{G_m} \omega_{K,\Lambda}$$

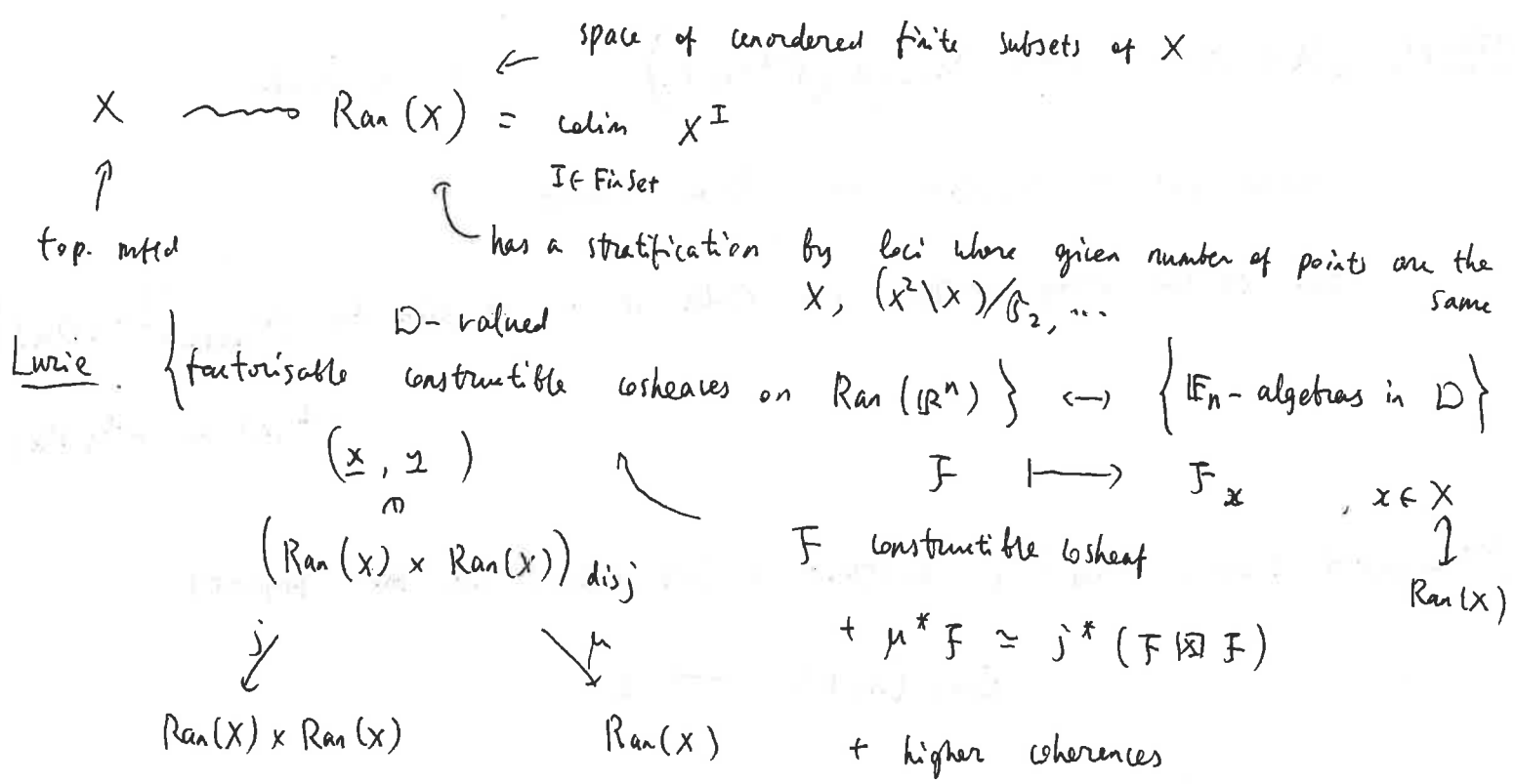
$$\text{Kir}(\omega_{K,\Lambda}) = \text{Shv}_{\text{exp}}(X)$$

Th. $\mathrm{K}^{\mathrm{B}}(\mathrm{Gr}_n) \simeq \mathrm{IndPerf}(\mathrm{B}\check{G})$

(Cass - van der Horst - Scholbach, ...)

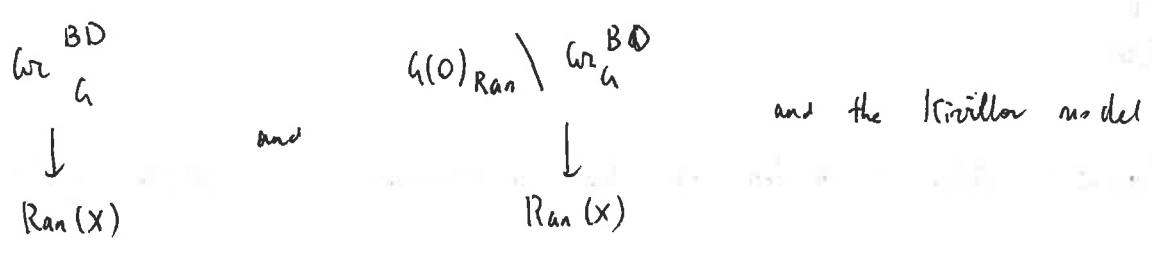


1. Idea. (Gaitsgory - Lurie) use Beilinson - Drinfeld Grassmannian to produce the extra IE_2 -structure.



$$F_{(\underline{x}, \underline{y})} \simeq F_{\underline{x}} \otimes_{\mathbb{D}} F_{\underline{y}}$$

If X is a complex curve (eg. $X = \mathbb{A}^1 \simeq \mathbb{R}^2$), there is a factorizable space



Task. Given a (suff. nice) map of stratified topological spaces

$$(Y, \mathcal{Q})$$

$\downarrow$

$$(X, P)$$

produce a const. cosheaf of categories on X

st. its fibre over $x \in X$ is given by

$$\text{Shv}_{\text{const}}(f^{-1}(x)).$$

Attempt. $U \subset X \rightsquigarrow \text{Shv}_{\text{const}}(f^{-1}(U))$ is a sheaf.

Could pass to adjoints $\leftarrow$ is a cosheaf

But it has wrong stalks. $\leftarrow$ stalk at x is given by $\text{Shv}_{\text{const}}(f^{-1}(D_x))$
 $f^{-1}(x) \hookrightarrow f^{-1}(D_x)$

(Kjærsgaard, Nowra, Wang) : construct a lax cosheaf with this property

$$\text{Enter}(X, P) \longrightarrow D$$

$$x_1 \rightsquigarrow x_2 \rightsquigarrow x_3 \rightsquigarrow$$

$$\begin{array}{ccc} 2. & \text{for } \begin{array}{c} \text{Gr}^{BD} \\ \text{Ran} \end{array} & \\ & \downarrow & \\ & \text{Ran} & \end{array}$$

this lax-cosheaf is strict.

3. (Prakhodko, Trautman) Also a version of this construction for Kirillov model action.

$$\text{Kir}(\text{Gr}_n, Sp) \bigotimes_{\mathbb{Z}} \mathbb{Z} \simeq \text{IndPerf}(B\check{U})$$